Sustainability of the Net International Investment Position (NIIP)

1MT404 Financial Stability

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Outline

1. Why the NIIP Matters
2. Concepts and Accounting
3. The NIIP Sustainability Model
4. Valuation in Practice: The United States
5. The Exorbitant Privilege
6. Advanced vs. Emerging Economies
7. The Czech Republic
8. Conclusions

Why Analyse the NIIP?

- **Gross positions have outgrown GDP:** US foreign assets are about 140% of GDP and liabilities about 210% (2026); Czech assets and liabilities each exceed 100% of GDP
- **Stocks, not only flows:** the current account (CA) is a flow; the NIIP is the stock it builds – and the stock also moves with prices and exchange rates
- **A vulnerability indicator:** the euro-area programme countries all entered the 2010–12 crisis with net positions far below -35% of GDP; the composition of liabilities (debt vs equity, currency, maturity) decided how hard they were hit
- **A policy variable:** the EU Macroeconomic Imbalance Procedure flags a NIIP below -35% of GDP; the IMF assesses external sustainability every year; the ČNB reports on it annually

Takeaway: Lecture 3 asked when flows become dangerous. This lecture asks when the *stock* they leave behind is sustainable.

The NIIP: Definition (IMF BPM6, Chapter 7)

Definition

The international investment position (IIP) shows, at a point in time, the value and composition of residents' financial **assets** that are claims on non-residents (plus gold bullion held as reserve assets) and residents' **liabilities** to non-residents:

$$NIIP_t = A_t - L_t$$

- **A stock:** the balance of payments records flows during a period; the IIP is the balance sheet at its end
- **Market valuation:** positions should be valued at market prices, converted at the end-period exchange rate
- **Link to national accounts:** the NIIP is the external part of the national balance sheet – the economy's net worth vis-à-vis the rest of the world
- **Sign:** $NIIP > 0$: net creditor (Germany, Japan, Norway); $NIIP < 0$: net debtor (US, Spain, Czech Republic)

Notation: A and L are stocks of foreign assets and liabilities; FA is reserved for the financial account (a flow). NFA (net foreign assets) $\equiv NIIP$.

The NIIP in the National Balance Sheet

Balance sheet of a sector

Assets	Liabilities and net worth
Real assets (land, machinery, inventories)	Liabilities to residents
Claims on residents (equity, other)	Liabilities to non-residents
Claims on non-residents, SDRs, monetary gold	Net worth

Consolidated balance sheet of the economy

Assets	Liabilities and net worth
Real assets	Liabilities to non-residents (debt and equity)
Claims on non-residents, SDRs, monetary gold	Net worth

- Consolidating all sectors cancels claims among residents: what one resident owes another is the other's asset
- What is left: **national net worth = real assets + NIIP**

Takeaway: A negative NIIP means that part of the domestic capital stock is owned or financed by non-residents – not that the country is "poor".

Source: IMF (1996), Balance of Payments Textbook, Illustrations 13.1 and 13.2 (simplified).

Functional Categories: Risk and Return

Category	What it is	Risk for the debtor country
Direct investment (FDI)	Lasting interest, at least 10% of voting power	Lowest: equity shares losses; returns fall in bad times; hard to withdraw
Portfolio equity	Shares below 10%	Shares losses, but prices can fall fast
Portfolio debt	Bonds and notes	Rollover risk; currency risk if in foreign currency
Other investment	Loans, deposits, trade credit	Highest: short-term bank funding runs first
Financial derivatives	Contracts for hedging or speculation	Margin calls in stress
Reserve assets	FX assets controlled by the central bank	An asset: the buffer against stops

Takeaway: The same NIIP can be safe or dangerous: equity liabilities share losses with foreign owners, debt liabilities do not. In the Czech Republic, direct investment is about 59% of all liabilities.

Source: IMF, BPM6, Chapter 6; Czech share: ČNB, investment position commentary (end-Q1 2024).

Balance of Payments Identities: A Recap

From lecture 3 (BPM6 sign convention)

$$CA + KA + NEO = FA$$

FA = net acquisition of foreign assets − net incurrence of foreign liabilities (net lending). A CA deficit is a net capital inflow: $FA < 0$.

Current account components:

$$CA = NX + PI + SI$$

NX net exports of goods and services; PI primary income (investment income and compensation of employees); SI secondary income (transfers)

Saving and investment (GNI basis):

$$CA = S - I$$

Reserves: with $FA = FA^{\text{non-res}} + \Delta RA$,

$$CA + KA + NEO - FA^{\text{non-res}} = \Delta RA$$

Takeaway: Identities, not causality: a deficit means domestic investment exceeds national saving and the gap is financed from abroad.

From Flows to Stocks: The IIP Identity

The change in the position (IMF)

$$NIIP_t - NIIP_{t-1} = \underbrace{FA_t}_{\text{transactions}} + \underbrace{OC_t}_{\text{other changes}}$$

Other changes are not in the balance of payments:

- **Price changes:** movements in equity and bond prices
- **Exchange-rate changes:** revaluation of foreign-currency positions into domestic currency
- **Other volume changes:** write-offs, reclassifications, changes in residence

Substituting $FA = CA + KA + NEO$:

$$NIIP_t = NIIP_{t-1} + CA_t + KA_t + NEO_t + OC_t$$

Takeaway: The NIIP equals cumulated current accounts only if prices, exchange rates and statistics never change.

A Worked Reconciliation: How the Position Changes in One Year

	End-1994	Transactions	Prices	Exch. rates	Other	End-1995
Assets	5,910	−879	+279	+1,071	+59	6,440
Direct investment	1,100	+106	+94	+203	+35	1,538
Portfolio investment	850	+328	+78	+162	−35	1,383
Other investment	2,320	−1,345	–	+384	−13	1,346
Reserve assets	1,640	+32	+107	+322	+72	2,173
Liabilities	5,801	+737	−33	+350	+9	6,864
Direct investment	3,104	−158	−45	+95	−16	2,980
Portfolio investment	252	+29	+12	+13	+18	324
Other investment	2,445	+866	–	+242	+7	3,560
NIIP	**109**	**−1,616**	**+312**	**+721**	**+50**	**−424**

Transactions: + = increase in the asset or liability (not the BoP sign convention).

Takeaway: Transactions cut the NIIP by 1,616, but price and exchange-rate gains of 1,033 offset almost two-thirds of it – the position fell by only 533.

Source: IMF (1996), Balance of Payments Textbook, Table 13.1 (hypothetical country "Essendonia").

Gross Positions and Valuation Effects

- Gross assets and liabilities grow regardless of net flows, so valuation acts on a large base:

$$VA_t \approx \pi_t^A A_{t-1} - \pi_t^L L_{t-1}$$

  with π^A, π^L the percentage change in the domestic-currency value of assets and liabilities (prices and exchange rates)
- **Illustration (US):** if two-thirds of US foreign assets ($47 trillion, mid-2026) are in foreign currencies, a 10% dollar depreciation raises their dollar value by about $3 trillion – about 10% of GDP and nearly three years of current account deficits
- **When valuation matters** (Atkeson, Heathcote and Perri 2025):
  1. gross positions are large relative to GDP
  2. the values of assets and liabilities do not move together (different currencies, different asset classes)

Takeaway: A small percentage revaluation of a large gross position can exceed a year of current account flows.

Source: U.S. Bureau of Economic Analysis, IIP Q2 2026; illustrative currency share.

Measuring the NIIP: Data, Dark Matter and Revisions

- **The database:** Lane and Milesi-Ferretti (2001, 2007) built the *External Wealth of Nations* – first 66 countries (1970–98), then 145 – and showed that cumulated current accounts fail to track net positions where gross positions are large
- **"Dark matter":** Hausmann and Sturzenegger (2007) capitalised US net investment income at a common yield and argued that the "true" US position was positive; critics replied that this only restates the excess return on US assets
- **Revisions can be huge:** BEA's June 2026 annual update revised the US position at end-2025 from $-\$27.5$ to $-\$21.9$ trillion – about 18% of GDP – mainly by lowering foreign-owned US liabilities after a new benchmark survey of foreign direct investment
- **Why:** FDI and unlisted equity have no market price; their values are estimated

Takeaway: The NIIP is an estimate, not an observation: compare positions within one data vintage and treat FDI values with caution.

Source: Lane and Milesi-Ferretti (2001, 2007); Hausmann and Sturzenegger (2007); BEA, U.S. international transactions and IIP, 1st quarter 2026 and annual update (24 June 2026).

The Model: Objective

Question: how does the NIIP evolve over time, and when is it sustainable in the long run? (Brůna 2013)

We proceed in four steps:

1. The return on the existing position, α_t
2. The dynamic equation: the position next year
3. Normalisation by GDP
4. The stabilising external balance

Key intuition: the NIIP changes through

- the external balance *before* investment income (trade, transfers, capital account)
- the return on the existing position – income, capital gains, exchange-rate revaluation
- GDP growth, which dilutes the position relative to the economy

Step 1: The Return on the External Position

The existing position NFA_{t-1} earns or costs, as a share of the position:

$$r_t = ir_t + div_t + rp_t + cg_t$$

interest, dividends, reinvested earnings and capital gains. Then the whole value is revalued by the exchange-rate change er_t:

$$\alpha_t = r_t + er_t\,(1 + r_t)$$

Sign rule:

- $\alpha_t NFA_{t-1} > 0$ adds to net wealth
- For a **debtor** ($NFA < 0$), $\alpha > 0$ is a **cost**

Example: $NFA_{t-1} = -100$, $r = 5\%$, foreign-currency position revalued by $er = +10\%$:

$$\alpha = 0.05 + 0.10 \times 1.05 = 15.5\%$$

The position worsens by 15.5 with no new transactions.

Takeaway: Even without any trade, the NIIP moves with yields, asset prices and the exchange rate.

Step 2: The Dynamic Equation – and the Double-Counting Trap

Split the current account: $CA_t = \underbrace{NX_t + CE_t + SI_t}_{\text{non-investment}} + IB_t$, with IB_t net investment income. Define the **external balance before investment income**:

$$B_t = NX_t + CE_t + SI_t + KA_t \ (+NEO_t)$$

Investment income and revaluation are both in $\alpha_t NFA_{t-1}$, so

$$\boxed{NFA_t = (1 + \alpha_t)\, NFA_{t-1} + B_t}$$

Trap

Writing $NFA_t = (1 + \alpha_t)NFA_{t-1} + CA_t$ counts interest, dividends and reinvested earnings twice – they are in the CA *and* in α. With the CA, use only the pure valuation part (cg and er).

Analogy: B plays the role of the primary fiscal balance in public debt dynamics; α the role of the interest rate.

Step 3: Normalisation by GDP

Define ratios to nominal GDP, $nfa_t = NFA_t/Y_t$ and $b_t = B_t/Y_t$, and nominal growth $Y_t = (1+\gamma_t)Y_{t-1}$.

Divide the dynamic equation by Y_t:

$$\frac{NFA_t}{Y_t} = (1+\alpha_t)\,\frac{NFA_{t-1}}{Y_{t-1}}\,\frac{Y_{t-1}}{Y_t} + \frac{B_t}{Y_t} \quad \Longrightarrow \quad \boxed{nfa_t = \frac{1+\alpha_t}{1+\gamma_t}\,nfa_{t-1} + b_t}$$

- $\alpha > \gamma$: the existing position *grows* relative to GDP unless b offsets it
- $\alpha < \gamma$: growth dilutes the position on its own

Takeaway: The same logic as public debt: what matters is the return on the position relative to growth, plus the balance before income.

Step 4: The Stabilising External Balance

Set $nfa_t = nfa_{t-1} = nfa$ and solve for b:

$$\boxed{b^* = \frac{\alpha - \gamma}{1 + \gamma}(-nfa)}$$

For a debtor ($nfa < 0$):

- $\alpha > \gamma$: the country must run a **surplus** before investment income ($b^* > 0$) – in practice a trade surplus
- $\alpha < \gamma$: it can run a permanent deficit ($b^* < 0$) and still keep nfa constant

In terms of the CA:

- With no valuation effects, a constant $nfa < 0$ needs a CA *deficit* of $\frac{\gamma}{1+\gamma}|nfa|$
- So "a debtor must run CA surpluses" is wrong – it must run surpluses in *trade*, large enough to pay the income

Takeaway: Brůna's long-run condition: the net export surplus must cover the net cost of the position minus growth dilution.

Worked Example: Czech Republic, 2025

% of GDP	2025
NIIP, end-2024 $\rightarrow$ end-2025	$-7.4 \rightarrow -9.8$
Net investment income	-4.6
Current account	$+0.7$
Capital account	$+1.5$
b = CA $-$ investment income + KA	$\approx +6.8$
Growth dilution, $\frac{\gamma}{1+\gamma}\lvert nfa\rvert$, $\gamma \approx 5\%$	$\approx +0.35$

- Transactions and growth: $+6.8 - 4.6 + 0.35 \approx +2.5$ pp
- Actual change: -2.4 pp
- Residual, about -5 pp: **revaluation** – the koruna appreciated in 2025, lowering the koruna value of the ČNB's reserves
- Stabilising balance: $b^* \approx (4.6 - 0.5)/1.05 \approx 4\%$ of GDP

Takeaway: The Czech trade surplus (about 5% of GDP) is what pays the income on foreign capital; the exchange rate then decides the year's result.

Source: ČNB, Zpráva o vývoji platební bilance 2025 (1 June 2026); calculations for this lecture.

Worked Example: United States, 2025–26

- $nfa \approx -70\%$ of GDP (mid-2026, revised data)
- Net primary income $\approx -0.15\%$ of GDP (2025): the position costs almost nothing
- Nominal growth $\gamma \approx 5\%$

$$b^* \approx \frac{0.15 - 0.05 \times 70}{1.05} \approx -3.2\% \text{ of GDP}$$

- The US can run a deficit before investment income of about 3% of GDP forever at a constant nfa
- In 2025: CA -3.6% of GDP, so $b \approx -3.45\%$ – close to b^*
- **Sensitivity:** if the US paid 1 pp more on its net position, b^* would shrink to about -2.5%

Takeaway: US sustainability rests on cheap liabilities: the risk is not the size of the position but a rise in what the US pays on it.

Source: BEA, international transactions (Q4 2025, Q2 2026) and IIP (Q2 2026).

Why One α Can Mislead: Gross Returns

The return on the net position hides two different returns:

$$\alpha_t \, NFA_{t-1} = \alpha_t^A A_{t-1} - \alpha_t^L L_{t-1}$$

United States

- $L/A \approx 1.5$ and net income ≈ 0
- So the US earns about 1.5 times the yield on its assets that it pays on its liabilities
- Safe debt sold, risky equity and FDI bought

Czech Republic

- Assets: ČNB reserves – safe, low yield
- Liabilities: FDI – risky, high return
- Net investment income -4.6% of GDP with a NIIP of only -10%: the "net α" would be over 40%

Takeaway: The cost of a position depends on what is on each side of the balance sheet, not on the net number.

Convergence to a Safe Level

To *improve* the position by at least δ pp of GDP a year (e.g. towards the EU MIP threshold of -35%):

$$\Delta nfa_t = b_t - \frac{\alpha_t - \gamma_t}{1+\gamma_t}\,|nfa_{t-1}| \;\geq\; \delta \quad\Longleftrightarrow\quad b_t \;\geq\; \delta + \frac{\alpha_t - \gamma_t}{1+\gamma_t}\,|nfa_{t-1}|$$

Example: from -80% to -35% in ten years ($\delta = 4.5$ pp) with $\alpha - \gamma = 1$ pp:

$$b \geq 4.5 + 0.01/1.01 \times 80 \approx 5.3\% \text{ of GDP every year}$$

- That is the scale of the adjustment Portugal, Spain and Greece faced after 2010
- A **sudden stop** forces it at once – through an import collapse rather than an export boom

Takeaway: Reducing a large negative position takes a decade of trade surpluses; markets may not wait that long.

Economic Intuition of the Model

What determines NIIP sustainability?

$$b^* = \frac{\alpha - \gamma}{1 + \gamma}(-nfa)$$

The external balance before income (b) must cover the cost of the position net of growth.

Three mechanisms:

1. The **external balance before income** (b) creates new claims or debt
2. The **return on the position** (α) – income and revaluation – changes its value
3. **GDP growth** (γ) dilutes the position relative to the economy

A negative NIIP is not necessarily a problem. It depends on:

- the currency of liabilities (own currency vs foreign currency)
- the instrument (FDI and equity vs debt)
- the gap between the cost of the position and growth

Gross External Financing Needs

Sustainability is a long-run question; **liquidity** is a short-run one. Each year a country must find:

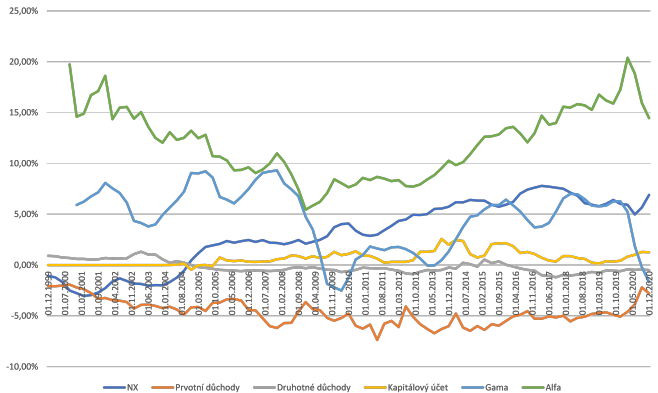
$$GEFN_t = \underbrace{-CA_t}_{\text{CA deficit}} + \underbrace{AM_t}_{\substack{\text{external debt falling due} \\ \text{within 12 months}}}$$

- The cost of servicing the position is already in the CA (investment income)
- AM = short-term debt by remaining maturity + amortisation of long-term debt
- **Greenspan–Guidotti rule:** reserves should at least cover external debt falling due within a year; the IMF's ARA metric adds portfolio liabilities, broad money and exports
- ČNB reserves were about 2.5 times the level needed to prevent and resolve a crisis (end-2025)

Takeaway: A sustainable position can still face a liquidity crisis if too much debt falls due at once.

Source: IMF external sustainability framework; ČNB, Zpráva o vývoji platební bilance 2025.

Sustainability Conditions: Czech Application, 1999–2020



Read-off:

- α (green) stays far above γ (light blue) in almost every year
- Primary income (orange) costs 4–7% of GDP
- Since 2005 net exports (dark blue) have covered it

Legend: NX = net exports; Prvotní důchody = primary income; Druhotné důchody = secondary income; Kapitálový účet = capital account; Gama = nominal GDP growth (γ); Alfa = return on the net position (α). % of GDP, four-quarter sums.

Source: Calculations following Brůna (2013); data: ČNB, ČSÚ.

Atkeson, Heathcote and Perri (2025): Decomposing the US NIIP

From $NFA_t - NFA_{t-1} = CA_t + VA_t + RES_t$, summing from $t = 1$ to T:

$$NFA_T = NFA_0 + \sum_{j=1}^{T} CA_j + \sum_{j=1}^{T} VA_j + \sum_{j=1}^{T} RES_j$$

- VA: valuation changes (prices and exchange rates); RES: other changes and statistical discrepancies
- **Finding:** the US position fell from about -10% of GDP in 2007 to about -65% in 2022, mainly because of a **US-specific rise in corporate asset values** – foreigners own a large share of US equity
- **Welfare:** rising US asset prices transferred wealth to foreign owners of US firms – the "end of privilege"
- Their charts scale positions by US corporate gross value added, the base for corporate valuations

Data in their paper predate BEA's June 2026 revision.

The US Net Foreign Asset Position, 1990–2022



Figure 1: The U.S. Net Foreign Asset Position: 1990-2022

Takeaway: Gradual decline to about -20% of GDP by 2002, a recovery to -4% in 2007, then a collapse to about -70% by 2021.

Source: Atkeson, Heathcote and Perri (2025), working-paper version (2023), Figure 1.

Decomposing Changes in US Net Foreign Assets

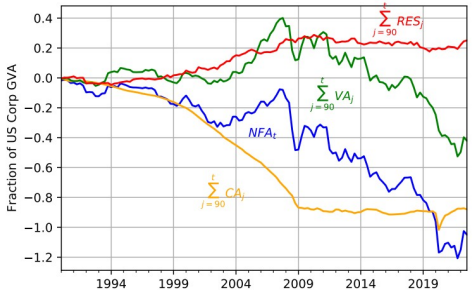


Figure 3: Decomposition of Changes in U.S. Net Foreign Assets over U.S. Corporate Value Added

Takeaway: Since 2007 cumulated valuation (green) fell from $+0.4$ to about -0.5 of corporate value added while cumulated current accounts (orange) moved by only about -0.3: valuation explains most of the decline.

Source: Atkeson, Heathcote and Perri (2025).

Gross Equity and Non-equity Positions

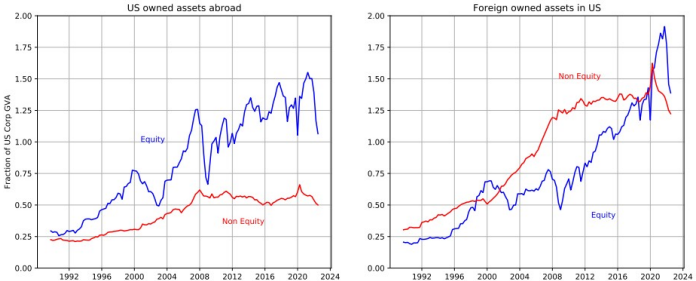
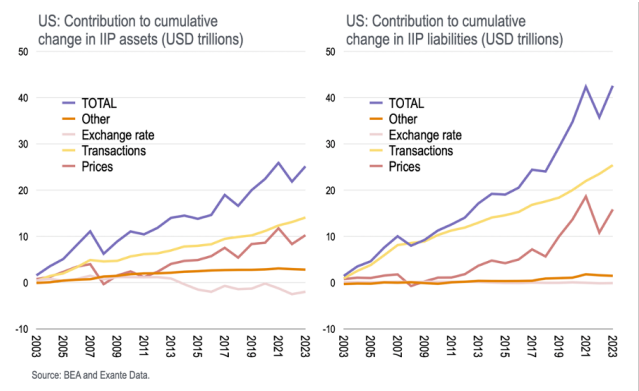


Figure 4: Gross Equity and Non-equity Positions over U.S. Corporate Value Added

Takeaway: US equity liabilities (right, blue) rose from about 0.5 to 1.9 times corporate value added after 2009, faster than US equity assets abroad (left, blue) – the US stock boom accrued partly to foreigners.

Source: Atkeson, Heathcote and Perri (2025), working-paper version (2023), Figure 4.

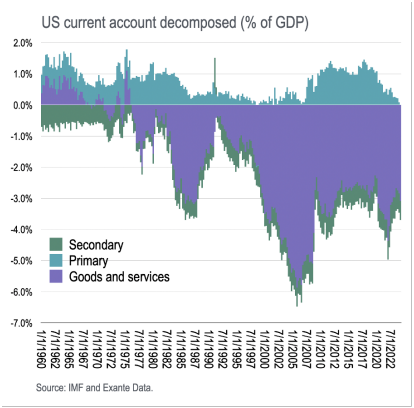
Contributions to Changes in US Assets and Liabilities, 2003–2023



Takeaway: Price changes added about $10 trillion to US assets but about $16 trillion to liabilities; exchange rates subtracted from assets after 2014 as the dollar strengthened.

Source: Exante Data (J. Nordvig), Money Inside Out (Substack), "United States: changing net IIP"; data: BEA (before the June 2026 revision).

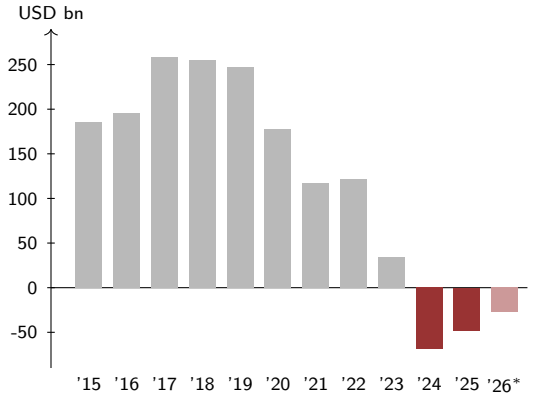
The US Current Account Decomposed



Takeaway: Primary income (light blue) offset part of the goods and services deficit for decades, but shrank to almost zero by 2024.

Source: Money Inside Out (Substack); data: IMF.

The Income Privilege Has Faded



US balance on primary income, USD billion (*first half of 2026)

- The surplus peaked at about $258 bn in 2017
- It turned negative in Q4 2023 and has stayed negative in all but one quarter since: $-\$69$ bn in 2024, $-\$48$ bn in 2025
- Higher interest rates raised payments on US debt liabilities faster than receipts on US assets

Takeaway: The US no longer earns more on its foreign assets than it pays on its liabilities; what remains of the privilege is cheap funding and valuation.

Source: U.S. Bureau of Economic Analysis, international transactions; series IEABCPI via FRED (updated 24 September 2026).

Comparison of Methods: Brůna vs. Atkeson, Heathcote and Perri

	Brůna (2013)	**Atkeson, Heathcote and Perri (2025)**
Main question	Is a transition economy's negative NIIP sustainable?	Why did the US position deteriorate after 2007, and who lost?
Key variable	Net return α against growth γ	Valuation of equity positions
Role of FX	Central for a small open economy	Secondary to equity prices for the US
Residual	Errors and omissions	Other changes and statistical discrepancies
Policy reading	Trade surplus needed to pay FDI income	Asset-price booms can transfer wealth abroad

Takeaway: Small open economies worry about the exchange rate; the US worries about its own stock market.

The Exorbitant Privilege

Definition

The term is attributed to French finance minister Valéry Giscard d'Estaing in the 1960s: the US can run persistent deficits without an explosive rise in its negative NIIP, thanks to the asymmetric composition of its balance sheet.

- **Liabilities:** mostly dollar-denominated and safe (Treasuries, deposits) – low yields
- **Assets:** largely in foreign currencies and risky (FDI, equity) – higher returns
- **Two channels:**
  - *Income:* the yield gap – for decades a primary income surplus despite a negative NIIP; negative since Q4 2023
  - *Valuation:* a weaker dollar raises the dollar value of assets, liabilities unchanged

Takeaway: The privilege is cheap funding plus insurance from the exchange rate – not a free lunch.

The Dollar Trap (Prasad 2014)

- The world needs safe dollar assets: central banks, firms and investors hold US Treasuries for liquidity and insurance
- Steady official and private demand keeps the yield on US liabilities low
- The US effectively **exports safety** and imports goods – trade deficits are the counterpart
- **The trap:** even as US debt grows, there is no alternative safe asset of comparable size (the euro area lacks a common one; the renminbi is not convertible), so demand persists

Takeaway: The dollar's role keeps US funding cheap – which is why the cost of the US position stays low (the US worked example).

Source: Prasad, E. (2014), The Dollar Trap: How the U.S. Dollar Tightened Its Grip on Global Finance, Princeton University Press.

The US as a "World Venture Capitalist"

Gourinchas and Rey (2007):

- Liabilities: short-term, cheap, safe debt
- Assets: long-term risky capital – FDI, equity, technology
- The US has earned an excess return on its foreign assets over its liabilities – the "world banker" turned "world venture capitalist"

The paradox today (*Money Inside Out*):

- Deeply negative position: about -70% of GDP
- Until 2023 the US still received more in dividends and interest than it paid
- Since then the income gap has closed (previous section)

Implication for stability: a negative NIIP can be sustained for a long time while the dollar remains the reserve currency – but a venture capitalist also takes the losses when global equities fall, and its funding cost can rise.

See lecture 3, slide 30, for the US IIP by component.

Source: Gourinchas and Rey (2007); Exante Data, Money Inside Out (Substack).

Advanced Economies

- **Currency structure:** liabilities largely in own currency – no exchange-rate risk in debt service
- **No "original sin":** they can borrow abroad in their own currency (Eichengreen and Hausmann 1999)
- **The central bank** can lower the domestic yield curve (QE), which lowers the cost of liabilities (α^L)
- **Safe-haven paradox:** in a crisis the domestic currency often appreciates (Switzerland, Japan). For a creditor with foreign-currency assets this *lowers* the NIIP in domestic currency – valuation losses exactly when risk rises

Takeaway: For advanced economies the exchange rate is a hedge or a source of creditor losses, rarely a debt problem.

Emerging and Transition Economies

Emerging markets (historically also the Czech Republic, Poland and Hungary) face specific constraints:

- **Net recipients of FDI:** stable funding, but structural dividend outflows on the current account
- **Original sin:** many cannot borrow abroad in their own currency – a depreciation raises the debt burden. The Czech Republic no longer suffers from it: non-residents hold koruna bonds
- **FX reserves:** a large share of assets earns low yields, while liabilities (FDI, private debt) cost more – a negative net income even with a small NIIP
- **Convergence:** real appreciation (Balassa–Samuelson) lowers the domestic-currency value of foreign-currency debt but also of foreign-currency assets such as reserves

Takeaway: For emerging economies, composition and currency matter more than the size of the net position.

How Policymakers Assess External Positions

Institution	Tool	What it flags
European Commission	Macroeconomic Imbalance Procedure (MIP) scoreboard	NIIP below -35% of GDP; three-year average CA outside -4% to $+6\%$; also NIIP excluding non-defaultable instruments (NENDI) and country-specific NIIP benchmarks
IMF	External Sector Report; External Balance Assessment (EBA)	CA gaps relative to a norm; the NIIP-stabilising CA; reserve adequacy (ARA metric)
ČNB	Annual balance of payments report	NIIP by sector and instrument, NIIP excluding FDI, reserves against crisis needs

Takeaway: Every framework looks beyond the headline NIIP – to instruments that can default, to the CA needed to stabilise the position, and to reserves.

Source: European Commission, MIP scoreboard; IMF, External Sector Report; ČNB, balance of payments reports.

Czech Republic: NIIP and Macro Context, 2023–2025

% of GDP	2023	2024	2025
CA	..	$+1.7$	$+0.7$
CA + KA	..	$+3.3$	$+2.2$
NIIP	-14.5	-7.4	-9.8

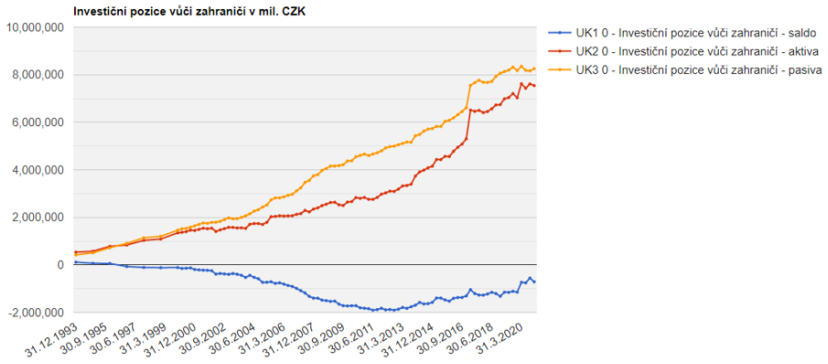
.. = not shown. 2023 NIIP as first published.

- **2024:** the best NIIP since 1999 – a large trade surplus (cars, recovering supply chains) plus revaluation: the koruna weakened, raising the koruna value of the ČNB's reserves and of foreign shares held by residents
- **2025:** the surplus shrank and the koruna appreciated – revaluation outweighed the surplus and the NIIP worsened
- Net investment income: -4.6% of GDP (2025), mostly FDI income

Takeaway: The Czech NIIP moves with the koruna as much as with the current account.

Source: ČNB, Zpráva o vývoji platební bilance 2024 (2 June 2025) and 2025 (1 June 2026).

Czech Gross Assets and Liabilities, 1993–2021

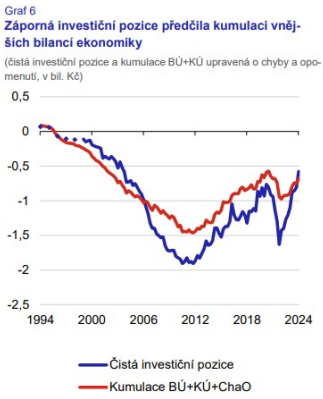


Legend: saldo = net position (blue); aktiva = assets (red); pasiva = liabilities (orange). CZK million.

Takeaway: Gross assets and liabilities rose from about CZK 0.5 trillion to over CZK 7.5 trillion; both jump in early 2017 – reserves on the asset side, non-residents' koruna holdings on the liability side – at the end of the koruna commitment. The net position bottomed at about −CZK 1.9 trillion around 2012.

Source: ČNB, ARAD database.

The NIIP and Cumulated External Balances



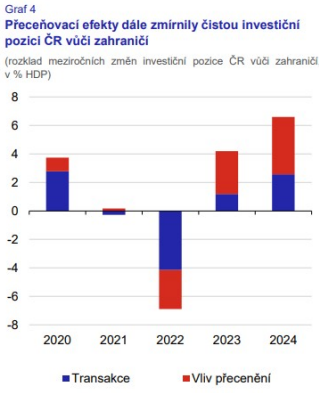
Read-off (CZK trillion):

- Blue: the NIIP. Red: cumulated CA + KA + errors and omissions – what the NIIP would be without revaluation
- From about 2000 to 2020 the NIIP was *below* the cumulated balances: revaluation worked against the country (rising value of foreign-owned Czech firms, koruna appreciation lowering foreign assets)
- In 2024 the NIIP rose above the cumulated balances for the first time in almost 20 years

Legend: Čistá investiční pozice = NIIP; Kumulace BÚ+KÚ+ChaO = cumulated current account + capital account + errors and omissions.

Source: ČNB, Zpráva o vývoji platební bilance 2024.

Revaluation Effects on the Czech NIIP



Year-on-year change in the NIIP, % of GDP:

- 2022: transactions -4.1 and revaluation -2.7 – the energy shock
- 2023 and 2024: revaluation added about $+3$ and $+4$ pp – more than transactions
- 2025 (not shown): the koruna appreciated and revaluation *lowered* the NIIP, outweighing the surplus

The weaker koruna in 2024 raised the koruna value of the ČNB's reserves by hundreds of billions – without a single extra export.

Legend: Transakce = transactions; Vliv přecenění = revaluation effect.

Source: ČNB, Zpráva o vývoji platební bilance 2024; 2025 report.

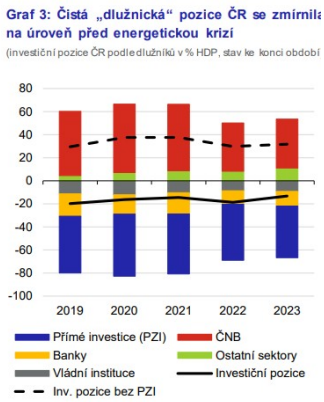
Who Is the Creditor and Who Is the Debtor?

Sector	Net position	Why
ČNB	**Strongly positive**	FX reserves, largely accumulated during the koruna commitment (2013–2017)
Government	**Negative**	Koruna government bonds held by non-residents (25% in 2025)
Banks	**Negative**	Funding from foreign parents; short-term non-resident deposits
Non-financial corporations	**Deeply negative**	Dominated by inward FDI (equity)
Households	**Positive**	Foreign shares and investment funds

Takeaway: The central bank and households are the creditors; firms owned by foreigners are the debtors – through equity, not debt.

Source: ČNB, balance of payments reports 2023–2025.

Investment Position by Debtor



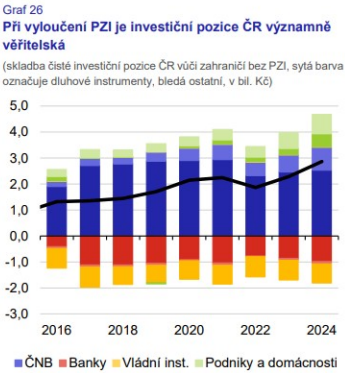
Read-off (% of GDP, end of year):

- The ČNB (red) is the main creditor: about $+40$ to $+60\%$ of GDP
- Direct investment (blue) is the main liability: about -45 to -55%
- Banks (orange) and government (grey) are smaller debtors
- The NIIP (solid line) is about -15 to -20%; excluding FDI (dashed) it is about $+30$ to $+38\%$

Legend: Přímé investice (PZI) = direct investment; Banky = banks; Vládní instituce = government; Ostatní sektory = other sectors; Inv. pozice bez PZI = NIIP excluding FDI; Investiční pozice = NIIP.

Source: ČNB, ČSÚ; ČNB, Zpráva o vývoji platební bilance 2023.

The NIIP Excluding FDI: A Net Creditor

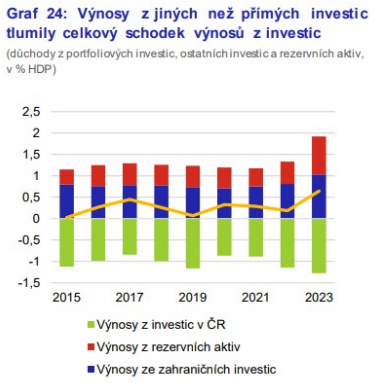


- Excluding FDI, the Czech Republic is a net creditor of about CZK 2.9 trillion – roughly a third of GDP (2024)
- The ČNB (blue) carries the creditor position; banks (red) and government (orange) are debtors
- **Currency structure:** foreign assets are in euros and dollars; liabilities are mostly in koruna
- So a weaker koruna *improves* the NIIP – the opposite of an emerging market with original sin

Legend: ČNB; Banky = banks; Vládní inst. = government; Podniky a domácnosti = corporations and households. Dark shade = debt instruments, light = other. CZK trillion.

Source: ČNB, Zpráva o vývoji platební bilance 2024.

Investment Income and the Interest Differential



Zdroj: ČNB, výpočet ČNB

Income from non-FDI investment, % of GDP:

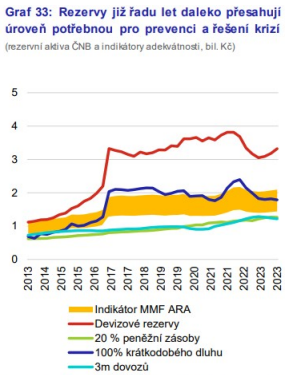
- Payments to non-residents (green): about -1 to -1.3%
- Receipts on foreign investment (blue) and reserves (red) rose sharply in 2023
- Net (yellow line): about $+0.6\%$ in 2023, up almost 0.5 pp

Why: the ČNB held its rate at 7% from mid-2022 and began cutting in December 2023; the ECB and the Fed kept raising rates until the second half of 2023. Yields on foreign-currency assets caught up, while the cost of koruna liabilities had already peaked.

Legend: Výnosy z investic v ČR = income paid on investment in the CR; Výnosy z rezervních aktiv = income on reserve assets; Výnosy ze zahraničních investic = income on foreign investment.

Source: ČNB, Zpráva o vývoji platební bilance 2023.

ČNB Reserves and Reserve Adequacy



Read-off (CZK trillion):

- Reserves (red) jumped from about 2.2 to 3.3 trillion in early 2017 – the last months of the koruna commitment
- Since then they have been well above every adequacy benchmark: the IMF ARA range (orange band), 100% of short-term debt (dark blue), 20% of broad money (green), three months of imports (cyan)
- End-2025: CZK 3.6 trillion (42.5% of GDP), about 2.5 times the level needed to prevent and resolve a crisis

Trade-off: the reserves are the main creditor asset – and the main source of revaluation swings.

Legend: Indikátor MMF ARA = IMF ARA metric; Devizové rezervy = FX reserves; peněžní zásoby = broad money; krátkodobého dluhu = short-term debt; dovozů = imports.

Source: ČNB, Zpráva o vývoji platební bilance 2023; 2025 report.

Foreign Direct Investment



Read-off (left axis: stocks, % of GDP; right axis: income flows):

- FDI in the Czech Republic (blue): about 75–80% of GDP throughout
- Czech FDI abroad (red): up from about 23% (2016) to about 32% (2024)
- FDI income paid abroad (light blue line): about 7–8% of GDP; received (orange line): about 1–2%

Czech firms are starting to invest abroad, but the stock of inward FDI is still more than twice as large.

Legend: PZI v ČR = FDI in the CR; PZI v zahraničí = Czech FDI abroad; Odliv výnosů PZI z ČR = FDI income outflow (right axis); Příliv výnosů PZI ze zahraničí = FDI income inflow (right axis).

Source: ČNB, Zpráva o vývoji platební bilance 2024.

FDI by Ultimate Owner and "Round-Tripping"

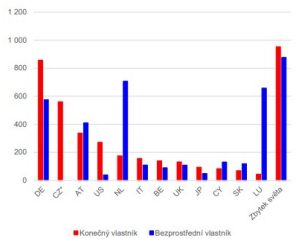
Ultimate investor

The *immediate* investor is the first foreign owner – often a holding in the Netherlands or Cyprus; the *ultimate* investor controls it.

- **Round-tripping:** domestic capital leaves for a foreign holding and returns as "FDI" (taxes, legal protection, anonymity)
- Czech residents are the second-largest ultimate owner of "foreign" FDI in the CR (left chart)
- Their share, about 14.5% (2019), is the highest among reporting countries (right chart)

Graf VI-2.1: Nejvíc v ČR investují Němci a Rakušané, nikoliv státy Beneluxu

(Přímé zahraniční investice v ČR podle země konečného a bezprostředního vlastníka, stav v roce 2019, mld. Kč)

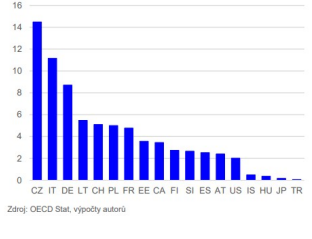


(*)Konečným vlastníkem je buď entita z ČR, nebo žádný zahraniční investor nedisponuje většinovou kontrolou.

Zdroj: ČNB, výpočty autorů

Graf VI-2.3: Podíl zahraničních investic s tuzemským konečným vlastníkem je v ČR nejvyšší mezi zeměmi, které tato data zveřejňují

(podíl PZI s vrcholovým investorem v domácí ekonomice v roce 2019 v %)



Zdroj: OECD Stat, výpočty autorů

Takeaway: Measured FDI liabilities overstate foreign ownership: part of the Czech "debt" is owed to Czechs.

Source: ČNB; OECD.Stat.

The Dividend Outflow Problem

- The large stock of FDI from the 1990s and 2000s generates large income for non-residents: net FDI income outflow of about 5.2% of GDP in 2025
- Two forms:
  1. **Paid dividends:** a cash outflow, recorded in primary income
  2. **Reinvested earnings:** stay in the country, but are recorded twice – as income paid (worsening the CA) and as a new FDI inflow (in the financial account)
- So the Czech CA understates the cash the economy keeps: part of the "outflow" is reinvested at home
- **Risk-sharing:** FDI income falls when Czech firms do badly – in a recession the outflow shrinks automatically, unlike interest on debt

Takeaway: The dividend outflow is the price of equity financing: costly in good times, a shock absorber in bad times.

Source: ČNB, Zpráva o vývoji platební bilance 2025.

Is the Czech NIIP a Risk? A Checklist

Dimension	Czech position	Assessment
Size	About -10% of GDP; MIP threshold -35%	Low
Instrument	Liabilities dominated by FDI equity	Low – risk-sharing
Currency	Assets in FX, liabilities mostly in koruna	Low – depreciation helps
Liquidity	Reserves about 2.5 times crisis needs	Low
Short-term funding	Non-resident deposits in banks (net inflow of 2.8% of GDP in 2025)	Watch – can run
Income	Net investment income -4.6% of GDP	Watch – needs a trade surplus

Takeaway: The Czech position is safe by composition, not by size: the things to watch are short-term non-resident deposits and the trade surplus that pays the FDI income.

Source: ČNB, Zpráva o vývoji platební bilance 2025; assessment for this lecture.

Conclusions

- **Stock, not flow:** the NIIP changes through transactions *and* revaluation; with large gross positions, revaluation can dominate – and statistics can be revised by trillions
- **The sustainability condition:** the external balance before investment income must cover the cost of the position net of growth, $b^* = \frac{\alpha-\gamma}{1+\gamma}(-nfa)$ – not "CA surpluses"
- **Composition beats size:** the currency, instrument and maturity of liabilities decide whether a negative NIIP is a vulnerability
- **The US:** sustainable while its funding stays cheap; the income privilege has gone, the valuation channel remains
- **The Czech Republic:** a small, equity-financed debtor with large reserves – safe by composition, with the trade surplus paying FDI income and the koruna driving annual swings

References (1/2)

- Atkeson, A., Heathcote, J., & Perri, F. (2025). The end of privilege: A reexamination of the net foreign asset position of the United States. *American Economic Review*, 115(7), 2151–2206.
- Brůna, K. (2013). Koncept udržitelnosti negativní čisté investiční pozice a jeho aplikace na příkladu České republiky v letech 1999–2011. *Politická ekonomie*, 61(1), 67–90.
- Bureau of Economic Analysis (2026). U.S. international transactions and investment position, 1st quarter 2026 and annual update (24 June); 2nd quarter 2026 (September).
- Česká národní banka (2024, 2025, 2026). *Zpráva o vývoji platební bilance* 2023, 2024, 2025.
- Eichengreen, B., & Hausmann, R. (1999). Exchange rates and financial fragility. NBER Working Paper 7418.
- European Commission. Macroeconomic Imbalance Procedure scoreboard.
- Gourinchas, P.-O., & Rey, H. (2007). From world banker to world venture capitalist: US external adjustment and the exorbitant privilege. In R. Clarida (ed.), *G7 Current Account Imbalances*. University of Chicago Press.
- Hausmann, R., & Sturzenegger, F. (2007). The missing dark matter in the wealth of nations and its implications for global imbalances. *Economic Policy*, 22(51), 469–518.
- IMF (1996). *Balance of Payments Textbook*. Washington, DC.
- IMF (2009). *Balance of Payments and International Investment Position Manual*, 6th ed. (BPM6).
- Lane, P. R., & Milesi-Ferretti, G. M. (2001). The external wealth of nations: Measures of foreign assets and liabilities for industrial and developing countries. *Journal of International Economics*, 55(2), 263–294.

References (2/2)

- IMF. *External Sector Report* (annual).
- Lane, P. R., & Milesi-Ferretti, G. M. (2007). The external wealth of nations mark II. *Journal of International Economics*, 73(2), 223–250.
- Nordvig, J. (Exante Data). *Money Inside Out* (Substack). United States: changing net IIP.
- Prasad, E. (2014). *The Dollar Trap: How the U.S. Dollar Tightened Its Grip on Global Finance*. Princeton University Press.
- U.S. Bureau of Economic Analysis, via FRED: series IIPUSNETIQ (net international investment position) and IEABCPI (balance on primary income).